

Standard Model of elementary particles

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Contents

1. Particle Zoo: Classification
2. Origins: cosmic rays and discoveries
3. Accelerators and Detectors
4. Fundamental Interactions
5. Standard Model formalism
6. Symmetries and Violations
7. The Higgs Boson
8. Beyond the Standard Model

1. Particle Zoo: Classification

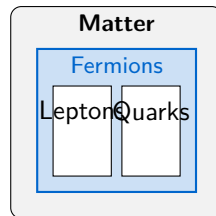
Classification Overview

Everything is classified by Spin and Interactions [11].

Type	Spin	Constituents	Force
Fermions	1/2, 3/2	Matter	-
Bosons	0, 1, 2	Forces	-
Hadrons	-	Quarks (q)	Strong
Leptons	1/2	Fundamental	Weak/EM

Hadrons Sub-groups

- **Baryons** (qqq): Fermions (e.g., Proton).
- **Mesons** ($q\bar{q}$): Bosons (e.g., Pion).



Historical Note: Heisenberg introduced **Isospin** to treat p and n as states of the same particle (Nucleon), differing only by an internal quantum number.

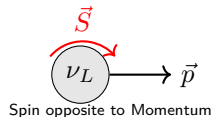
Leptons: Generation 1

Electron (e^-)

- **Mass:** 0.511 MeV.
- **Charge:** -1 .
- **Discovery:** J.J. Thomson (1897).
- **Role:** Chemistry, Electricity.
- **Stability:** $> 10^{26}$ years.

Electron Neutrino (ν_e)

- **Mass:** < 1 eV (almost zero).
- **Charge:** 0.
- **Interaction:** Only Weak.
- **Chirality:** Always Left-Handed.
- **Lepton Number (L):** Conserved.
 $\mu^- \rightarrow e^- \gamma$ is forbidden!

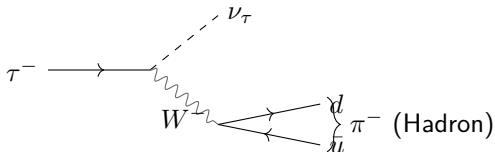


Leptons: Generations 2 and 3

Heavier than electrons. Unstable.

Particle	Mass	Charge	Lifetime	Decay Mode
Muon (μ^-)	106 MeV	-1	$2.2\mu s$	$\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu$ <i>Discovered in Cosmic Rays (1937). Penetrates matter deeply.</i>
Tau (τ^-)	1.77 GeV	-1	$10^{-13}s$	$\tau^- \rightarrow \text{hadrons} + \nu_\tau$ <i>Discovered at SLAC (1975). Heavy enough to create pions!</i>

Example: Tau decay to Hadrons:



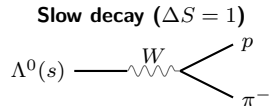
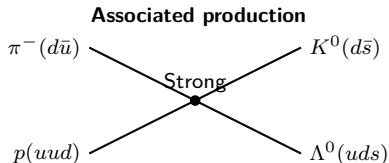
Quarks: Light (u, d) and Strange (s)

Quarks are fractionally charged fermions confined in Hadrons.

Flavour	Charge	Mass	Strangeness
u	$+2/3$	$2 - 3 \text{ MeV}$	0
d	$-1/3$	$4 - 6 \text{ MeV}$	0
s	$-1/3$	100 MeV	-1

Strangeness Paradox

- **Production:** Copious via Strong force (associated production: $\pi p \rightarrow K \Lambda$).
- **Decay:** Very slow via Weak force (10^{-10} s vs 10^{-23} s).



Quarks: Heavy (c, b, t)

- **Charm (c):** Mass 1.4 GeV. Charge $+2/3$.
 - "November Revolution" (1974): discovery of J/ψ .
 - Solved theoretical issues (GIM Mechanism).
- **Bottom (b):** Mass 4.5 GeV. Charge $-1/3$.
 - Discovery: 1977 (Υ meson). Key for CP Violation.

Top Quark (t)

Mass $\sim 172 - 173$ GeV (heavier than a Gold atom!). Charge $2/3$

- **Unique:** Decays ($t \rightarrow Wb$) **before** it can hadronize ($\tau \ll \tau_{QCD}$).
- It is the only "bare" quark we can study directly.

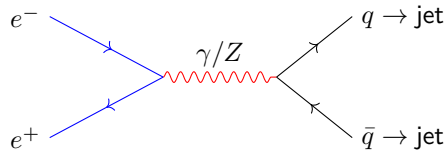
Evidence for Quarks: R-Ratio

Evidence for **Colour** charge ($N_c = 3$). Each quark flavour comes in three colours (red, green, blue).

$$R = \frac{\sigma(e^+e^- \rightarrow \text{Hadrons})}{\sigma(e^+e^- \rightarrow \mu^+\mu^-)} \propto N_c \sum Q_q^2$$

where Q_q is the electric charge of the quarks with mass less than the beam energy.

- Without colour ($N_c = 1$), R is too small.
- With $N_c = 3$, theory matches data.
- **Jets:** Quarks hadronize into sprays of particles.



Detecting unstable particles: Resonances

Short-lived particles (like Δ , J/ψ , Z) don't leave tracks. They are detected as peaks in the Invariant Mass spectrum.

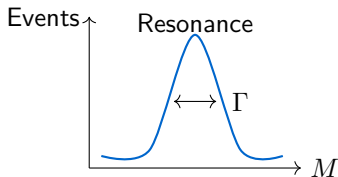
Heisenberg Uncertainty Principle:

$$\tau \cdot \Gamma \approx \hbar$$

- τ : Lifetime of the particle.
- Γ : Width of the mass peak (Energy uncertainty).

Rule:

- Short lifetime ($\tau \sim 10^{-23}s$) $\rightarrow$ Broad peak (Strong decay).
- Long lifetime ($\tau \sim 10^{-20}s$) $\rightarrow$ Narrow peak (EM/Weak decay).



Hadrons: Baryons (qqq) and Decuplet

Composite fermions (Spin 1/2 or 3/2). Color singlet (RGB).

Common Baryons (Spin 1/2):

- **Proton (p):** uud . Stable.
- **Neutron (n):** udd . Unstable.
- **Lambda (Λ^0):** uds .

Delta Paradox

$\Delta^{++}(uuu)$ has 3 identical fermions in same state (violating Pauli principle). Impossible!
→ The wavefunction becomes antisymmetric by introducing **Colours**.

Baryon Decuplet (Spin 3/2)

$$\Delta^{-} \quad \Delta^0 \quad \Delta^{+} \quad \Delta^{++} (uuu)$$

$$\Sigma^{*-} \quad \Sigma^{*0} \quad \Sigma^{*+}$$

$$\Xi^{*-} \quad \Xi^{*0}$$

$$\Omega^{-} \quad (sss)$$

Predicted by Gell-Mann, found in 1964.[3]

Mathematics of Forces: Yang-Mills

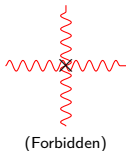
Forces are described by matrices rotating internal states (Gauge Theory).

- **QED (Quantum Electrodynamics) ($U(1)$):** 1×1 matrices (Phases). Commutative (Abelian). Electromagnetic forces between charged particles.
- **QCD (Quantum Chromodynamics) ($SU(3)$):** 3×3 matrices (Colours). Non-commutative (Non-Abelian). Strong force binding quarks.

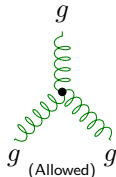
Non-Abelian Consequence

Unlike photons, Gluons carry colour charge. They interact with each other!

QED: No Photon Self-Int.



QCD: Gluon Vertex



Building Baryons: Spin Statistics

Using the "Lego block" analogy to combine 3 quarks (u, d, s).

Spin 3/2 (Decuplet)

- Spins aligned: $\uparrow\uparrow\uparrow$
- Symmetric in Space, Flavor, Spin.
- **Must** be antisymmetric in Color.
- Example: Ω^- ($s^\uparrow s^\uparrow s^\uparrow$).

$$\begin{array}{c} \uparrow \uparrow \uparrow \\ J = 3/2 \end{array}$$

Spin 1/2 (Octet)

- Mixed symmetry: $\uparrow\uparrow\downarrow$
- Example: Proton (uud).
- uuu configuration is forbidden here by Pauli principle (without color mixing).

$$\begin{array}{c} \uparrow \uparrow \downarrow \\ J = 1/2 \end{array}$$

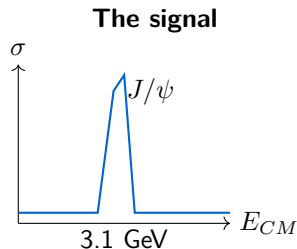
November Revolution (1974)

Discovery of J/ψ ($c\bar{c}$) simultaneously at SLAC and Brookhaven.

- **Observation:** A narrow resonance at 3.1 GeV (J/ψ).
- **Confirmation:** Proved the existence of the Charm quark ($c\bar{c}$).

GIM Mechanism (1970)

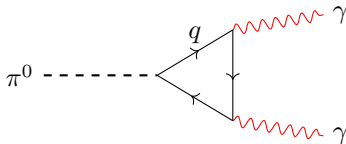
Glashow, Iliopoulos, Maiani predicted Charm to explain the suppression of $K^0 \rightarrow \mu^+ \mu^-$ (FCNC) [4].



Hadrons: Mesons ($q\bar{q}$)

Composite bosons. Mass carriers of nuclear force (Yukawa).

Meson	Quarks	Spin	Main Decay
Pion (π^+)	$u\bar{d}$	0	$\pi^+ \rightarrow \mu^+ \nu_\mu$ (Weak)
Pion (π^0)	$u\bar{u} - d\bar{d}$	0	$\pi^0 \rightarrow \gamma\gamma$ (EM, Fast)
Kaon (K^+)	$u\bar{s}$	0	Strange! Decays to pions.
J/ψ	$c\bar{c}$	1	$\rightarrow e^+e^-/\mu^+\mu^-$



2. Origins: cosmic rays and discoveries

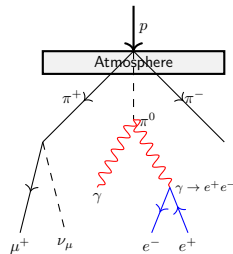
Pre-Accelerator era: Cosmic Rays

Before the 1950s, the universe was our only 'accelerator' [9, 5, 10].

Discovery (Hess, 1912)

Radiation increases with altitude. Confirmed extraterrestrial origin.

- **Primary Rays:** Protons (85%), α , Nuclei.
- **Process:** $p + \text{Atmosphere} \rightarrow \pi^{\pm,0} + X$.
- **Showers:** Pions decay into muons (μ) and neutrinos (ν).
- **Energy:** Up to 10^{20} eV (LHC $\approx 10^{13}$ eV).



Visualizing particles: Cloud Chambers

Cloud chamber

Supersaturated vapor condenses along the ionization trail of a charged particle.

1. Positron (e^+) [1]

- Particle with mass m_e curving the "wrong" way in B-field.
- First proof of Antimatter.

2. Muon (μ)

- Mass $200m_e$. Penetrates deep underground.
- Initially mistaken for Yukawa's meson (Strong force carrier).
- **Problem:** It didn't interact with nuclei! (Conversi-Pancini-Piccioni exp).

$\pi \rightarrow \mu$ Decay Chain

Powell (1947) used Nuclear Emulsions to distinguish Pion from Muon.

- **Pion (π^-):** Stops, then decays.

$$\pi^- \rightarrow \mu^- + \bar{\nu}_\mu$$

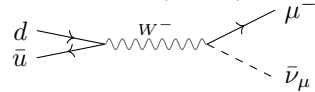
(2-body decay: Monochromatic μ)

- **Muon (μ^-):** Travels, then decays.

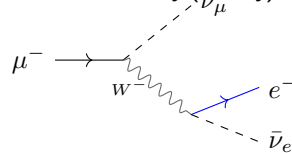
$$\mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu$$

(3-body decay: Spectrum of e^-)

1. Pion Decay (2-body)



2. Muon Decay (3-body)



Antimatter and Neutrino

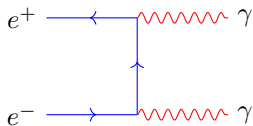
Dirac Equation (1928)

$$(i\gamma^\mu \partial_\mu - m)\psi = 0$$

Predicted negative energy states ("Holes") → **Antimatter**.

Annihilation $e^+e^- \rightarrow \gamma\gamma$

[Pair annihilation feynman diagram]

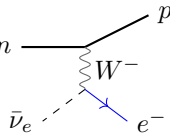


Beta Decay

$n \rightarrow p + e^-$ presented two problems:

- Continuous energy spectrum (Energy violation?).
- Spin: $1/2 \rightarrow 1/2 + 1/2$ (Angular momentum violation?).

Pauli's solution: Neutrino: neutral particle (Spin $1/2$). $n \rightarrow p + e^- + \bar{\nu}_e$



Neutrinos

Neutrinos are neutral, weakly interacting leptons. They come in 3 "Flavours".

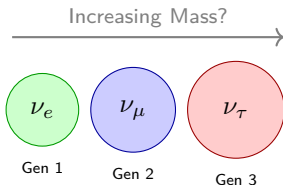
Key Properties:

- **Charge:** 0 (Electric neutral).
- **Spin:** $1/2$ (Fermions).
- **Mass:** Extremely small (< 1 eV), but $\neq 0$.
- **Interaction:** Only Weak (and Gravity).

Ghost Particle

Trillions of neutrinos pass through you every second without interacting. To stop a neutrino, you would need a lead wall 1 light-year thick!

Flavor	Sym	Partner	Discovery
Electron	ν_e	e^-	1956 (Reactor)
Muon	ν_μ	μ^-	1962 (Beam)
Tau	ν_τ	τ^-	2000 (DONUT)



3. Accelerators and Detectors

Accelerators

Why high energy?

1. **Probe small scales:** $\lambda = h/p$ (Microscope).
2. **Create heavy particles:** $E = mc^2$ (Synthesis).

Type	Mechanism	Limit
Cyclotron	Constant B, spiral	Relativistic mass desync.
Synchrotron	Varying B, fixed R	Synchrotron Rad ($\propto E^4/m^4$).

Energy: Collider vs Fixed target

To create a particle of mass M :

- **Collider:** $E_{beam} = M/2$ (Linear gain).
- **Fixed Target:** $E_{beam} \approx M^2/2m_T$ (Quadratic cost!).

Relativistic kinematics: Invariant Mass

How to find short-lived particles (Resonances) like $Z, H, J/\psi$?

Invariant Mass

For a decay $X \rightarrow 1 + 2$:

$$M^2 = (E_1 + E_2)^2 - \|\vec{p}_1 + \vec{p}_2\|^2$$

- Measure stable daughters (e.g., μ^+, μ^-).
- Plot Events vs Mass. A Peak indicates a particle.
- Width Γ is related to lifetime: $\tau = \hbar/\Gamma$.

Cross section and Luminosity

The "reaction rate" of particle physics.

- **Cross Section (σ):** Interaction probability (Area).
- **Luminosity ($\mathcal{L}$):** Beam intensity ($cm^{-2}s^{-1}$).

Total Events (N):

$$N = \sigma \cdot \int \mathcal{L} dt = \sigma \cdot L_{int}$$

Example: Higgs $\sigma \approx 50$ pb. With $L_{int} = 100 \text{ fb}^{-1}$, we expect 5 million Higgs.

Rare Events

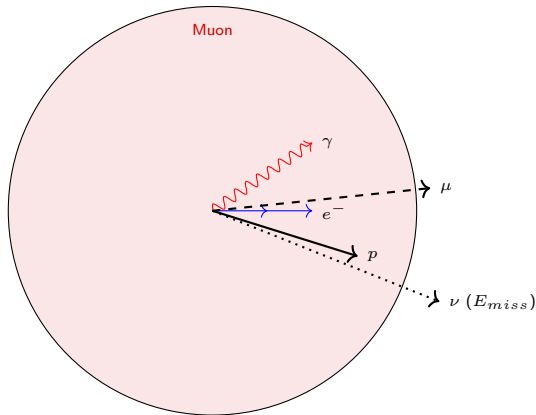
Higgs production has $\sigma \sim pb$. Requires massive $\mathcal{L}$ to see any events.

Modern detectors: onion structure

1. **Tracker:** Measures momentum of charged particles ($p = 0.3BR$).
2. **ECAL:** Lead/Crystals. Stops $e^{\pm}, \gamma$.
3. **HCAL:** Brass/Iron. Stops Hadrons (p, π, K).
4. **Muon Chambers:** Only muons penetrate here.

Problem

Neutrinos escape detection! **Missing transverse energy** (E_T^{miss}).



4. Fundamental Interactions

Overview of Forces

Interactions are mediated by the exchange of **Gauge Bosons** (Spin 1).

Force	Mediator	Mass (GeV)	Charge	Spin	Range	Couples to
Strong	Gluon (g)	0	0	1	10^{-15}m	Color Charge
EM	Photon (γ)	0	0	1	∞	Electric Charge
Weak	$W^{\pm}, Z^0$	80.4, 91.2	$+1/-1(W), 0(Z)$	1	10^{-18}m	Flavor/Isospin
Gravity	Graviton?	0	0	2	∞	Energy/Mass

Yukawa Principle

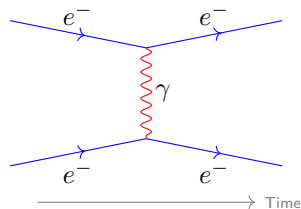
The range of a force is inversely proportional to the mass of its mediator: $R \sim \frac{\hbar}{Mc}$ EM/Gravity have infinite range because $M_{\gamma,g} = 0$. Weak is short-range because W, Z are heavy.

Electromagnetic Interaction (QED)

Quantum Electrodynamics: The most precise theory in physics.

- **Mediator:** Photon (γ). Massless, Charge 0.
- **Coupling:** $\alpha \approx 1/137$ (Fine Structure constant).
- **Vertex:** Fermion-Fermion-Photon. Conserves Flavor, Charge, Color.

Key Process: Møller Scattering ($e^-e^- \rightarrow e^-e^-$).



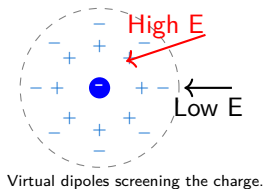
QED: Running Coupling (α)

Is the charge constant? No! It depends on the energy scale Q^2 [11].

- **Vacuum Polarization:** Virtual e^+e^- pairs from the vacuum screen the central charge.
- **Effect:** At higher energy (shorter distance), we penetrate this cloud and see a larger effective charge.

$$\alpha(M_Z) \approx 1/128 \quad (\text{vs } 1/137 \text{ at } Q^2 \approx 0)$$

Note: In QCD, the effect is opposite (α_s decreases with Energy). This suggests all forces might unify at 10^{16} GeV (GUT).

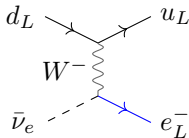


Weak Interaction: $W^\pm$ and Z^0

The only force that changes Flavour and violates Parity (P).

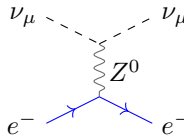
Charged Current ($W^\pm$)

- **Chirality:** Couples ONLY to **Left-Handed** fermions (u_L, e_L).
Right-handed particles don't feel it!
- **Example:** β -Decay ($d \rightarrow uW^-$).



Neutral Current (Z^0)

- No flavor change ($q \rightarrow qZ$).
- Couples to both L and R (but differently).
- **Example:** $\nu_\mu e^- \rightarrow \nu_\mu e^-$ (Elastic).



Electroweak Unification

Glashow, Weinberg, Salam (Nobel 1979). [12] EM and Weak are low-energy manifestations of a single gauge symmetry $SU(2)_L \times U(1)_Y$.

Mixing Mechanism

The physical photon (γ) and Z^0 are mixtures of the original neutral fields B^0 and W^3 :

$$\begin{pmatrix} \gamma \\ Z^0 \end{pmatrix} = \begin{pmatrix} \cos \theta_W & \sin \theta_W \\ -\sin \theta_W & \cos \theta_W \end{pmatrix} \begin{pmatrix} B^0 \\ W^3 \end{pmatrix}$$

where θ_W is the **Weinberg Angle** ($\sin^2 \theta_W \approx 0.23$).

- **Prediction:** Existence of Neutral Currents (Z^0) before discovery.
- **Mass Relation:** $M_W = M_Z \cos \theta_W$.

Weak interaction: Real vs Virtual

Virtual Mediators (Low E)

- In decays (e.g., muon decay), $Q^2 \ll M_W^2$.
- The propagator $1/(Q^2 - M_W^2) \approx -1/M_W^2$.
- Looks like a "contact force" (Fermi Theory).

Real Mediators (High E)

- Produced at colliders ($\sqrt{s} \geq M_{W,Z}$).
- Discovered at CERN (UA1/UA2) in 1983.
- **Z-Pole (LEP):** By counting Z decays, we proved there are exactly 3 light neutrino families.

Strong Interaction (QCD)

Quantum Chromodynamics: theory of quarks and gluons.

- **Charge:** "Color" (Red, Green, Blue).
- **Mediators:** 8 Gluons.
- **Composition:** They carry **Color + Anticolor** (e.g., $r\bar{g}, b\bar{r}$).
- **Non-Abelian:** Since they have color, gluons attract each other!

Confinement

Potential $V(r) \sim -\frac{\alpha}{r} + \kappa r$.

- Small r : Coulomb-like (Free quarks).
- Large r : Linear (String tension). Impossible to isolate a quark.



Unlike photons, gluons have charge.

QCD: Asymptotic Freedom

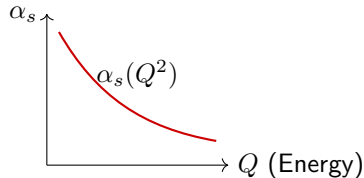
The opposite of QED screening!

- Vacuum Polarization in QCD: Gluon loops have an "anti-screening" effect.
- **Result:** The coupling α_s decreases at high energies.

High Energy $\rightarrow$ Weak Force

Consequence

Inside the proton, quarks move freely (Parton Model). But if you try to pull them apart (low energy), the force becomes infinite.



Nobel Prize 2004 (Gross, Politzer, Wilczek).

5. Standard Model formalism

Gauge Groups and Lagrangian

The Standard Model is a QFT based on the local gauge symmetry group:

$$G_{SM} = SU(3)_C \times SU(2)_L \times U(1)_Y$$

Group	Coupling	Bosons	Acts on...
$U(1)_Y$	g'	B^0 (1)	Hypercharge
$SU(2)_L$	g	W^1, W^2, W^3 (3)	Left-Handed Fermions
$SU(3)_C$	g_s	$g_1 \dots g_8$ (8)	Color Charge (Quarks)

Lagrangian

$$\mathcal{L}_{SM} = \underbrace{-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}}_{\text{Forces}} + \underbrace{i\bar{\psi}\not{D}\psi}_{\text{Matter}} + \underbrace{|D_{\mu}\phi|^2 - V(\phi)}_{\text{Higgs Mass}} - \underbrace{y\bar{\psi}_L\phi\psi_R}_{\text{Fermion Mass}}$$

Where: $F_{\mu\nu}$ represents all gauge tensors (G^a, W^i, B). The Covariant Derivative D_{μ} contains the interactions:

$$D_{\mu} = \partial_{\mu} - ig'\frac{Y}{2}B_{\mu} - ig\frac{\tau^i}{2}W_{\mu}^i - ig_s\frac{\lambda^a}{2}G_{\mu}^a$$

Standard Model Lagrangian ($\mathcal{L}_{SM}$)

The master equation describing all fundamental interactions (except gravity).

$$\mathcal{L}_{SM} = \underbrace{-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}}_{\text{I. Gauge}} + \underbrace{i\bar{\psi}\not{D}\psi}_{\text{II. Matter}} + \underbrace{|D_{\mu}\phi|^2 - V(\phi)}_{\text{III. Higgs}} - \underbrace{y_{ij}\bar{\psi}_L\phi\psi_R}_{\text{IV. Yukawa}}$$

Note: interaction vertex is a term from $\bar{\psi}\gamma^{\mu}D_{\mu}\psi$, i.g. $\bar{\psi}\gamma^{\mu}(g\tau W_{\mu})\psi$: fermion emitting W boson.

I. Gauge Sector (Kinetic): Describes the propagation of force carriers (g, W, B).

- $F_{\mu\nu} = \partial_{\mu}A_{\nu} - \partial_{\nu}A_{\mu} + g[\mathbf{A}^{\mu}, \mathbf{A}^{\nu}]$
- The commutator term $[A, A]$ exists only for Non-Abelian forces ($SU(2), SU(3)$).
- **Physics:** It allows Gluons and W bosons to interact with themselves (Self-interaction).

II. Matter Sector (Dirac): Describes the kinematics of Quarks and Leptons.

- $\not{D} = \gamma^{\mu}D_{\mu}$: Feynman slash notation.
- **Physics:** Matter fields ψ are massless here. They only acquire mass via interaction with the Higgs.

III. Higgs Sector (EWSB): Describes the scalar field ϕ and Spontaneous Symmetry Breaking.

- $|D_{\mu}\phi|^2$: Kinetic term. When $\phi \rightarrow v$, this generates mass terms for gauge bosons (W, Z).
- $V(\phi) = \mu^2|\phi|^2 + \lambda|\phi|^4$: The "Mexican Hat" potential responsible for the VEV.

IV. Yukawa Sector (Fermion Mass):

- Links Left (ψ_L , Doublet) and Right (ψ_R , Singlet) fields via the Higgs.
- **Physics:** After symmetry breaking ($\phi \rightarrow v$), this term becomes a mass term: $m_f = \frac{y_f v}{\sqrt{2}}$

Interaction mechanism: Covariant Derivative

To maintain Local Gauge Invariance $\psi \rightarrow e^{i\alpha(x)}\psi$, the standard derivative ∂_μ is replaced by D_μ .

Formula

$$D_\mu = \partial_\mu - ig' \frac{Y}{2} B_\mu - ig \frac{\tau^i}{2} W_\mu^i - ig_s \frac{\lambda^a}{2} G_\mu^a$$

1. Hypercharge ($U(1)_Y$)

- **Field:** B_μ (1 boson).
- **Generator:** Y (Scalar).
- **Action:** Simple phase rotation.
- **Physics:** Precursor to Electromagnetism.

2. Weak ($SU(2)_L$)

- **Field:** $W_\mu^{1,2,3}$ (3 bosons).
- **Generator:** τ^i (Pauli Matrices, 2×2).
- **Action:** Mixes Doublets.

$$\begin{pmatrix} \nu_e \\ e^- \end{pmatrix}_L \xrightarrow{W} \begin{pmatrix} e^- \\ \nu_e \end{pmatrix}_L$$

- **Physics:** Only force that changes particle flavor.

3. Strong ($SU(3)_C$)

- **Field:** $G_\mu^{1\dots 8}$ (8 Gluons).
- **Generator:** λ^a (Gell-Mann Matrices, 3×3).
- **Action:** Mixes Colors.

$$\begin{pmatrix} r \\ g \\ b \end{pmatrix} \xrightarrow{G} \begin{pmatrix} g \\ r \\ b \end{pmatrix}$$

Weinberg Rotation and EW unification

At high energy, EM and Weak are unified. We don't see γ and Z , but B and W^3 .

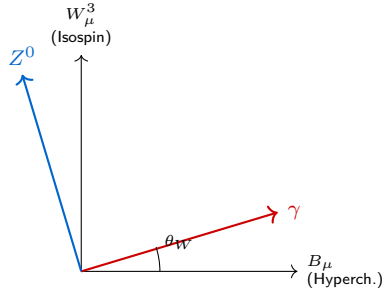
- **Weak Isospin (T):** Couples to W .
- **Hypercharge (Y):** Couples to B .
- **Relation:** $Q = T_3 + \frac{Y}{2}$.

Weinberg Rotation

Physical bosons are mixtures of gauge fields. The mixing angle θ_W is fixed by the ratio of coupling constants:

$$\tan \theta_W = \frac{g'}{g} \approx \frac{\text{EM strength}}{\text{Weak strength}}$$

$$\begin{pmatrix} \gamma \\ Z^0 \end{pmatrix} = \begin{pmatrix} \cos \theta_W & \sin \theta_W \\ -\sin \theta_W & \cos \theta_W \end{pmatrix} \begin{pmatrix} B \\ W^3 \end{pmatrix}$$



The Symmetry Breaking "chooses" a direction, creating the massless photon.

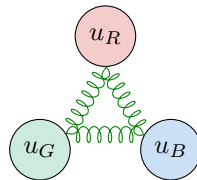
QCD and Color

Why do we need Color? Consider the Δ^{++} baryon (uuu) [9].

Problem:

- **Flavor:** uuu (Symmetric).
- **Spin:** $J = 3/2$ ($\uparrow\uparrow\uparrow$) (Symmetric).
- **Space:** $L = 0$ (Symmetric).

Pauli Principle: Fermion wavefunction must be **Antisymmetric!**



Δ^{++} (Colorless)

Solution: Colour

Introduce a new quantum number $SU(3)_C$. $\Psi = \psi(space) \cdot \psi(spin) \cdot \psi(flavor) \cdot \psi_{color}$ **Colour Confinement Hypothesis:** Only "Colourless" states (Singlets) are observable in nature.

- Baryons: $RGB = \text{White}$ (Antisymmetric ϵ_{ijk}).
- Mesons: $R\bar{R} = \text{White}$.

6. Symmetries and Violations

Discrete Symmetries (C, P, T)

- **Parity (P):** Spatial inversion.

$$(x, y, z) \rightarrow (-x, -y, -z)$$

- **Charge Conjugation (C):** Particle $\leftrightarrow$ Antiparticle.

$$e^- \rightarrow e^+$$

- **Time Reversal (T):**

$$t \rightarrow -t$$

CPT Theorem

Any Lorentz-invariant local QFT must be invariant under the combined operation CPT.

- Implications: Particle and Antiparticle must have exact same mass and lifetime.

Helicity and Chirality: V-A Theory

The Weak force is unique: it discriminates based on "Handedness" (Chirality).

- **Helicity:** $h = \frac{\vec{S} \cdot \vec{p}}{|\vec{p}|}$.
- **Under Parity:** $\vec{p} \rightarrow -\vec{p}$, but $\vec{S} \rightarrow \vec{S}$ (Axial Vector).

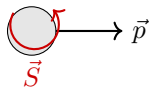
$$P(h) = -h$$

Parity flips Left $\leftrightarrow$ Right.

The V-A Interaction

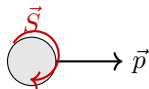
The W boson couples **only to Left-Handed** chiral states. Since the mirror reflection of a Left-state is a Right-state (which doesn't interact), **Parity is maximally violated**.

Left Handed



Spin opposite Momentum

Right Handed



Parity violation: Wu experiment (1957)

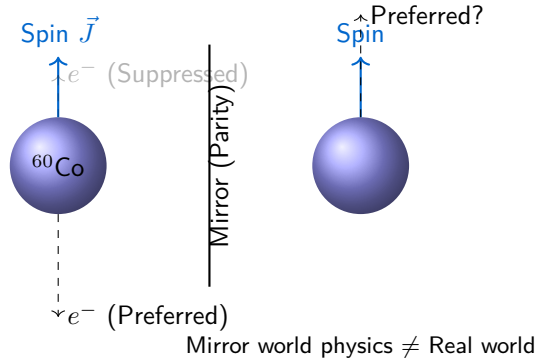
Experimental proof using Cobalt-60 (^{60}Co) decay [13].

Setup:

- ^{60}Co nuclei aligned by B-field (Spin $\uparrow$).
- Decay: $^{60}\text{Co} \rightarrow ^{60}\text{Ni} + e^- + \bar{\nu}_e$.

Observation: Electrons were emitted preferentially opposite to the spin direction.

Conclusion: If Parity were conserved, emission would be symmetric (Up = Down). It was not.



The CKM Matrix: Quark Mixing

Mass eigenstates $(d, s, b) \neq$ Weak eigenstates (d', s', b') . Quarks "mix" when interacting with the W boson.

$$\begin{pmatrix} d' \\ s' \\ b' \end{pmatrix} = \underbrace{\begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix}}_{\text{CKM Matrix}} \begin{pmatrix} d \\ s \\ b \end{pmatrix}$$

Hierarchy of Strength

$$\begin{pmatrix} \mathbf{0.97} & 0.22 & 0.004 \\ 0.22 & \mathbf{0.97} & 0.04 \\ 0.008 & 0.04 & \mathbf{0.99} \end{pmatrix}$$

Quarks prefer to stay in their generation ($V_{tb} \approx 1$).

- **Off-diagonal terms** allow flavor changes across generations.
- **Cabibbo Angle** (θ_c): $V_{us} \approx \sin \theta_c \approx 0.22$.

Why 3 Generations?

To have CP Violation, the matrix must be complex (imaginary phase).

- **2 Gens:** 1 Rotation angle (Cabibbo).
No Phase.
- **3 Gens:** 3 Angles + **1 Phase** (δ).

Kobayashi and Maskawa (1973) predicted the 3rd generation (b, t) *before* discovery just to allow CP violation!

CP Violation and Kaons

CP: Symmetry between Matter and Antimatter.

- **Expectation:** CP is conserved. K^0 (Matter) and $\bar{K}^0$ (Antimatter) should behave symmetrically.
- **Discovery (Cronin and Fitch, 1964):** The neutral Kaon K_L (mostly $CP = -1$) was found to decay into $\pi\pi$ ($CP = +1$).

$$K_L \rightarrow \pi\pi \quad (\text{Violation} \sim 0.2\%)$$

Cosmological Significance

CP violation is one of the 3 **Sakharov Conditions** needed to explain why the universe is made of Matter and not Antimatter.

(However, SM CP violation is too small to explain the full asymmetry!)

7. The Higgs Boson

Mass Problem

Conflict

- **Bosons:** Gauge symmetry forbids mass terms $m^2 W_\mu W^\mu$.
- **Fermions:** Chiral symmetry forbids mass terms $m \bar{\psi}_L \psi_R$ (since L is a doublet, R is a singlet).

Reality: W, Z , quarks and leptons are massive!

How to give mass to W/Z without breaking the renormalizability of the theory?

- **Solution: Spontaneous Symmetry Breaking (SSB).**
- The Lagrangian is symmetric, but the **Vacuum** (ground state) is not.

The Higgs Potential

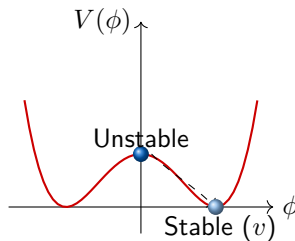
Introduce a complex scalar field ϕ with potential:

$$V(\phi) = \mu^2 |\phi|^2 + \lambda |\phi|^4$$

- If $\mu^2 < 0$, the minimum is at $v = \sqrt{-\mu^2/\lambda} \neq 0$.
- The field acquires a **Vacuum Expectation Value (VEV)**:

$$\langle \phi \rangle = v \approx 246 \text{ GeV}$$

- The symmetry "rolls" down to a random direction in the valley.



Generating Mass: Higgs Mechanism

Particles interact with the background Higgs field (v) which acts like a "cosmic molasses", slowing them down (giving inertia/mass). [6]

Symmetry Breaking Pattern

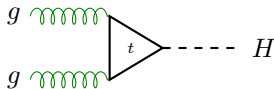
$$SU(2)_L \times U(1)_Y \rightarrow U(1)_{EM}$$

- 3 generators break $\rightarrow$ 3 Goldstone bosons eaten by $W^\pm, Z^0$ (Mass).
- **1 generator remains unbroken (Q) $\rightarrow$ Photon remains massless.**
- 1 massive scalar remains $\rightarrow$ **Higgs Boson (H).**

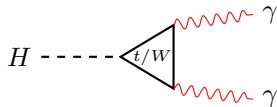
Fermion Mass: Acquired via Yukawa coupling ($f\bar{f}\phi$). $m_f \propto g_f \cdot v$ *Top quark is heavy because it couples strongly ($g_t \approx 1$) to the Higgs.*

Discovery at LHC (2012)

Production (Gluon Fusion): Does H couple to gluons? No (no color)! It interacts via a **Top Quark Loop**.



Decay ($H \rightarrow \gamma\gamma$): Proceeds via W or Top loop. Rare (BR $\sim 0.2\%$) but very clean signal.



Result

Mass $M_H \approx 125$ GeV. Spin 0 (Scalar).

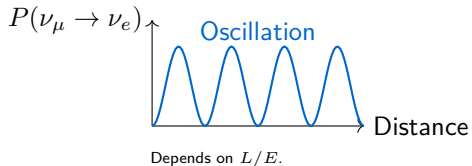
8. Beyond the Standard Model

Neutrino Oscillations

Neutrinos change flavor ($\nu_e \leftrightarrow \nu_\mu \leftrightarrow \nu_\tau$) as they travel [11, 2, 8].

- **Implication:** $\Delta m^2 \neq 0 \implies$ **Neutrinos have mass!**
- **Mixing:** Described by the **PMNS Matrix**.

$$\begin{pmatrix} \nu_e \\ \nu_\mu \\ \nu_\tau \end{pmatrix} = U_{PMNS} \begin{pmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{pmatrix}$$



Difference from CKM: Mixing angles are LARGE.

Example: ν_2 is a near-equal mix of all flavors.

Neutrino nature: Seesaw mechanism

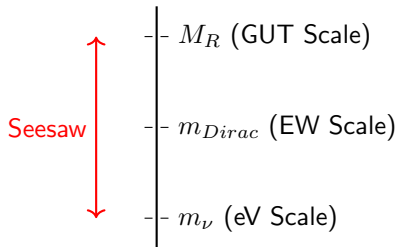
Neutrino masses are tiny (< 1 eV). Why?

Dirac Mass: Standard ($m\bar{\nu}_L\nu_R$). **Majorana Mass:** Particle = Antiparticle.

Type I Seesaw

Introduce a heavy Right-Handed Neutrino (N_R) with mass $M \sim 10^{15}$ GeV.

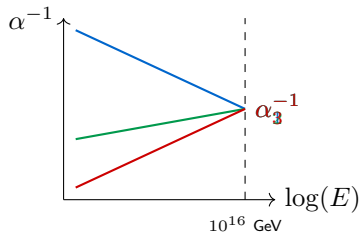
$$m_\nu \approx \frac{m_{Dirac}^2}{M}$$



Grand Unified Theories (GUTs)

Do all forces merge at high energy?

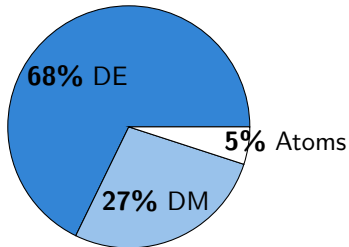
- **Running Couplings:** $\alpha_1, \alpha_2, \alpha_3$ evolve with energy.
- **Standard Model:** They nearly meet at 10^{15} GeV, but miss!
- **Supersymmetry (SUSY):** Modifies the slope, making them meet perfectly at $M_{GUT} \approx 2 \cdot 10^{16}$ GeV.
- **Prediction:** Proton Decay ($p \rightarrow e^+ \pi^0$). Not seen yet ($\tau_p > 10^{34}$ yrs).



The unknown universe

The Standard Model describes ordinary matter perfectly, but that's only part of the story. *"The Standard Model is the most successful theory in history, yet we know it must be wrong."*

- **Dark Matter (27%):** Invisible mass holding galaxies together (Rotational curves).
- **Dark Energy (68%):** Unknown energy driving the accelerated expansion.
- **SM Matter (5%):** Everything we have studied so far!



Theoretical 'puzzles':

- Hierarchy Problem ($M_H \ll M_{Pl}$).
- Matter/Antimatter Asymmetry.

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